

Chapter 5

Consumption and Investment

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Intertemporal Decisions

- › Households make **intertemporal decisions**
 - Decisions that affect variables across time (a type of dynamic decision)
 - Ex: Choosing how much to *save* so that you can consume today and in the future
- › Like with everything in economics, this comes with a price

The Interest Rate

- › The interest rate is the relative price of consumption today
 - If the rate is 4%, I can consume \$100 today, or \$104 next year
 - Impacts how much you choose to save
- › Interest on savings is the opportunity cost of consuming
- › Useless semantic point: the relative price of consuming today in this case is 1.04, which is $(1 + \text{interest rate})$. Saying just the interest rate is common shorthand.

Why Do Interest Rates Exist?

- › Lenders need to be compensated for:
 - Impatience: Rather have \$100 today or that same \$100 in a year?
 - Opportunity cost: What you would have done with that \$100
 - Risk: World could end tomorrow and you'll never see that \$100
 - Market power: Micro class :(

Individual Savings Decisions

- › Let's start with a basic *real* model of a savings choice:

- › HH makes choices in two periods:
 1. Today
 2. Tomorrow

- › They choose:
 1. Consumption today (c)
 2. Consumption tomorrow (c')
 3. How much to save for tomorrow (s)

Saving vs Borrowing

- › HH's save by lending to other HH's this period
- › Next period, borrowers pay back the loan
- › Similar to what happens by saving in a bank

Given Variables

- › The HH takes as given
 - Income today (y)
 - Income tomorrow (y')
 - Interest rate (r)
- › We've assumed an **endowment economy**
 - Income is given in both periods
- › The **endowment point (E)** will always be feasible
 - Meaning the budget line **MUST ALWAYS** go through this point
 - This income (y, y') is given to the consumer regardless

Intertemporal Budget Constraint

› Two budget constraints:

- $y = c + s$
- $y' + (1+r)s = c'$

› After some algebra we can create one budget constraint:

$$y + \frac{y'}{(1+r)} = c + \frac{c'}{(1+r)}$$

– We've eliminated the savings choice from the equation

Plot-able Budget Line

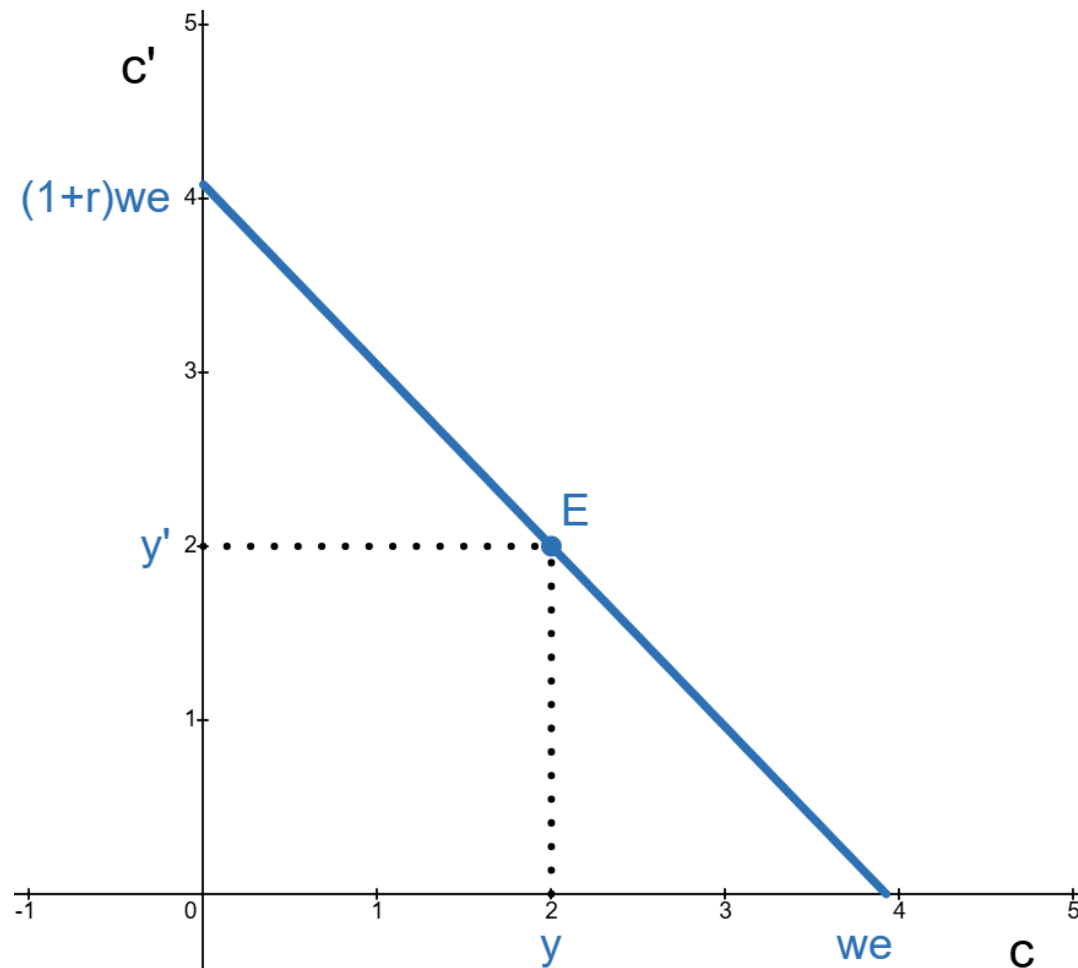
- › Now define: $w^e \equiv y + \frac{y'}{(1+r)}$
 - **Net present lifetime wealth:** total income *value* in both time periods
- › After some more algebra:

$$c' = -(1+r)c + (1+r)w^e$$

Analyzing the Budget Line

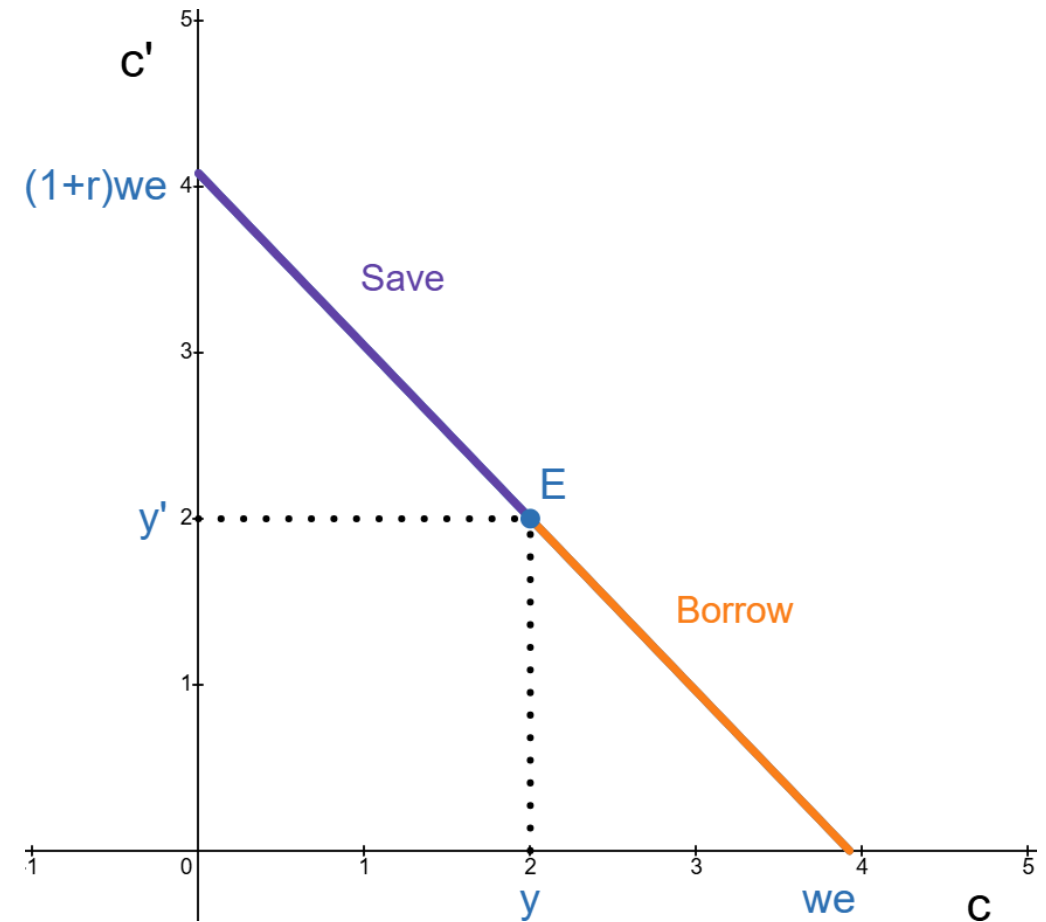
- $c' = -(1+r)c + (1+r)w^e$
- › Future consumption is the amount of wealth you didn't spend in the past, plus the interest from saving it
 - Consume nothing in the past: $c' = w^e(1+r) \rightarrow$ y-intercept
 - Making $-(1+r)$ the slope (note the negative sign)
 - ***WHY is the (abs. val.) of the slope (1+r)?***

Plotting in the (c, c') Space



To Borrow or Save?

- › Starting from what they're given (E), the HH can:
 - Save, consume more later
 - or
 - Borrow, consume more now
- › Why do we know the HH will be on a point on the budget line?



Unpacking HH Utility

- › We know the HH wants to maximize utility
 - Last ch. we didn't dive into the utility function
 - Now, we'll talk about the time component
- › Assume HHs value the future *less* than the present
 - Doesn't have to be by a lot; even just .01% less will work
 - › Prefer to have something today, or that same thing a year from now?
- › Define: $\beta \in [0, 1)$ as constant that expresses how much the HH values the future relative to the present

The Household's Problem

› The HH's problem is:

$$\max\{U(c, c')\} \text{ st: } w^e = c + \frac{c'}{(1+r)}$$

- Maximize utility from c and c' , subject to their budget
 - › Remember w^e is total lifetime wealth
- Where $U = u(c) + \beta u(c')$
- Notice there's our β placing a weight on the value of c'

Marginal Rate of Substitution

› Just as in the last chapter, we have a *MRS*:

$$MRS_{c,c'} = \frac{MU(c)}{\beta MU(c')} = (1+r)$$

- › Again, this is our HH market clearing condition
 - Interest must afford enough consumption tomorrow to make up for the utility lost by giving up consuming today
 - HH is indifferent between marginal increase in c or c'

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Proof (not required)

Consumption Choice of Saver (lender)

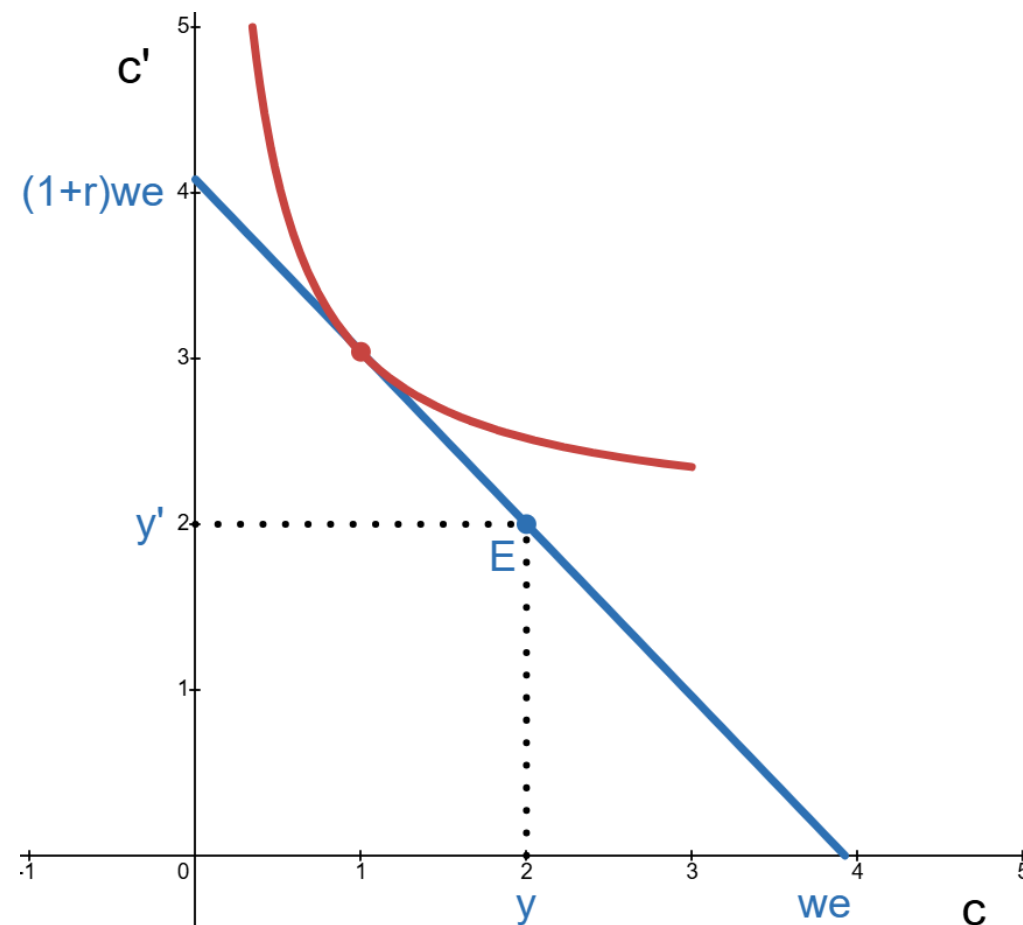
› Given this utility, HH picks:

- $c = 1$
- $c' = 3$

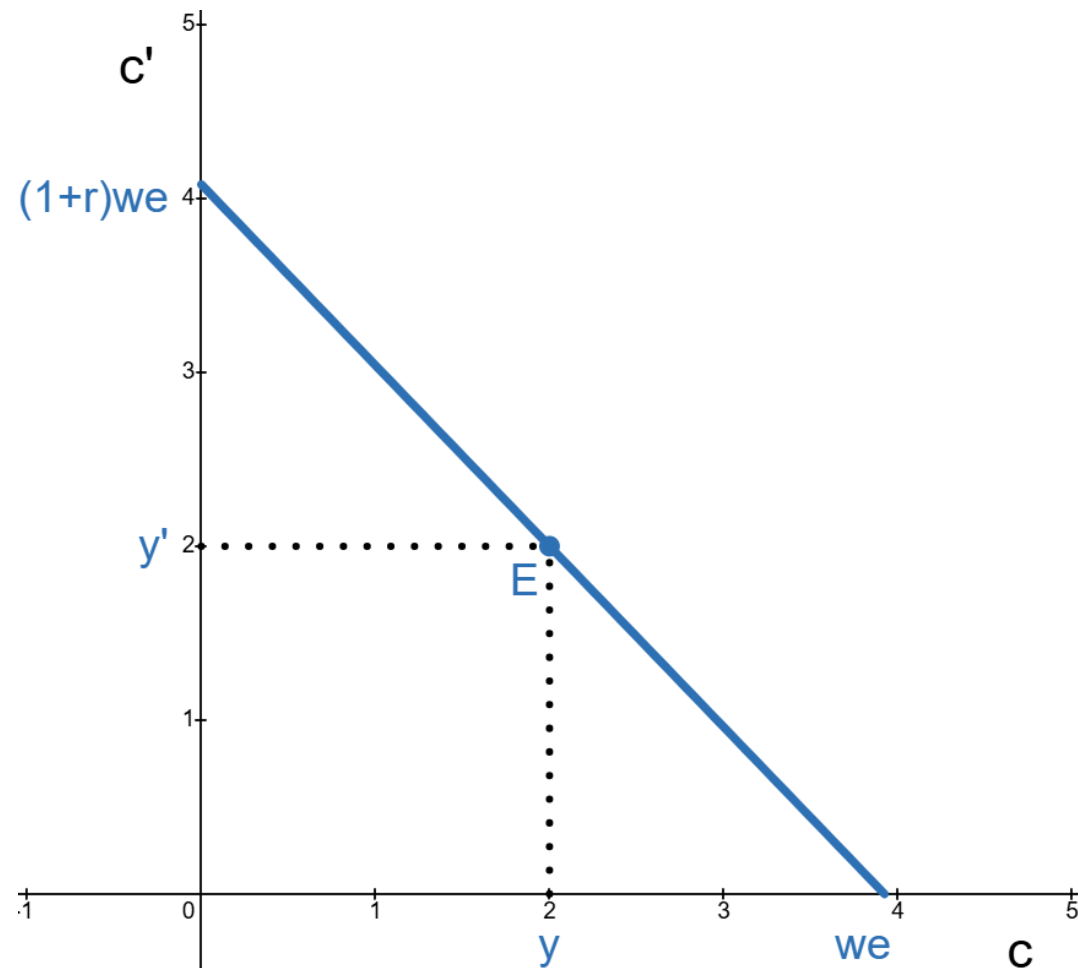
› Savings = 1

– Why?

› What is the slope of IC at (c, c') ?



What Does the Borrowing Case Look Like?



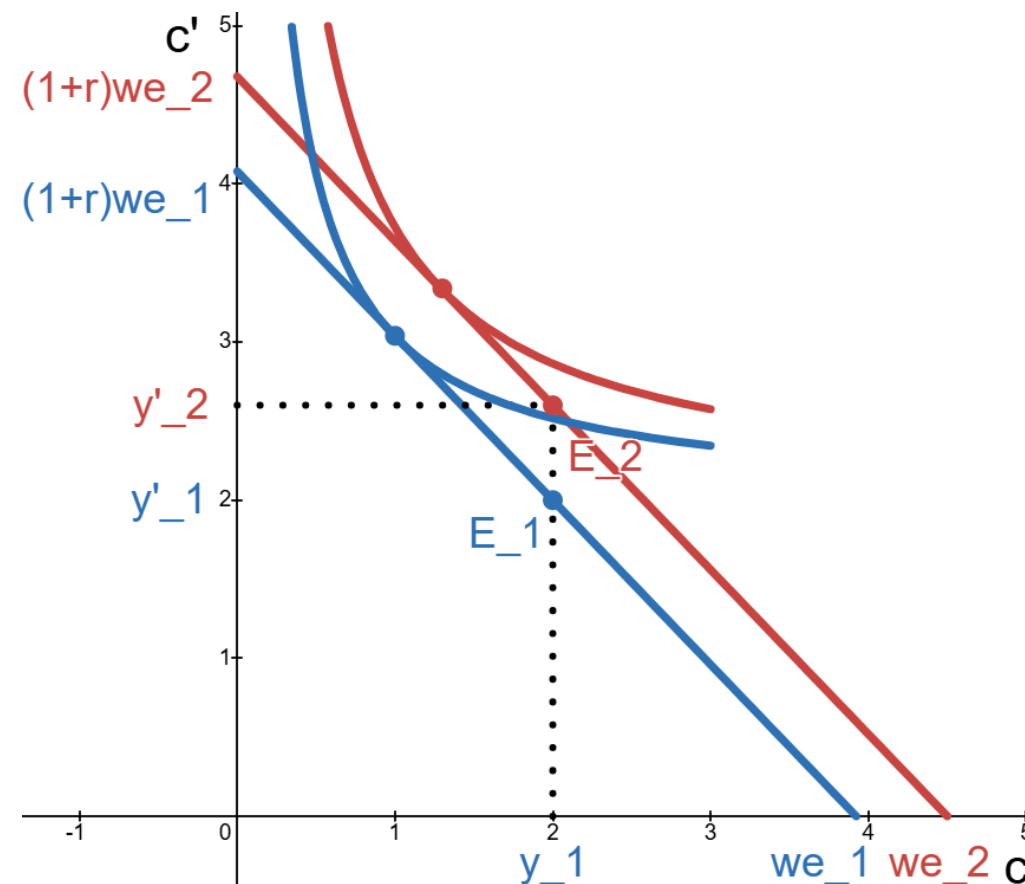
Change in Future Income

- › Assume the HH gets a boost in future income: $y'_1 \rightarrow y'_2$
 - Endowment increases
 - Budget must go through point new E' point

- › Wealth must also increase: $w^e_2 = y_1 + \frac{y'_2}{(1+r)}$

Graphing an Increase in Future Income

- › Wealth increased:
 - Both w^e and $(1+r)w^e$ must increase
- › No change in $r \rightarrow$ slope stays the same
- › Pure wealth effect



Wealth Effect from Change in Endowment

- › There is only a wealth effect in this case
 - No substitution effect
- › **Why?** the relative price, r , did not change

The substitution effect only occurs when the relative price changes. The relative price is the rate at which the HH *substitutes* one good for another

Change in Interest Rate: Two Cases

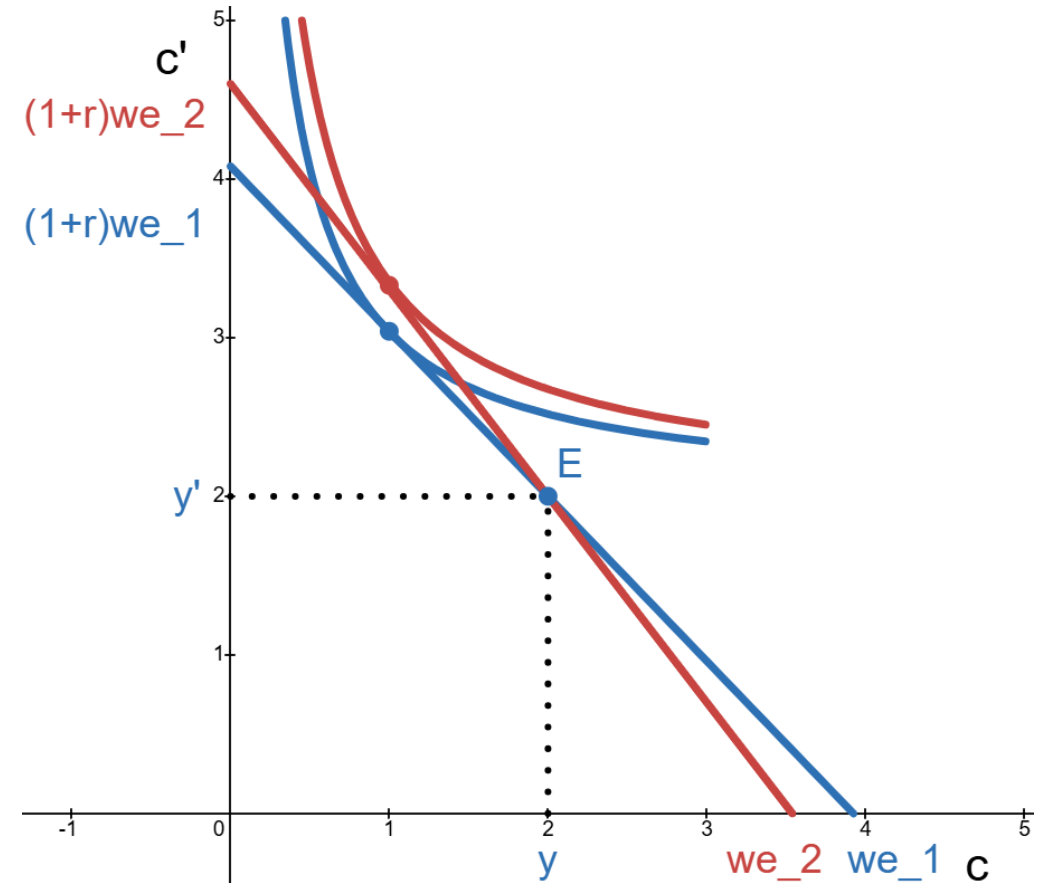
- › Recall the slope of the budget was $-(1+r)$
 - Changing r changes the slope
- › The endowment doesn't change
 - Budget must still go through point E

Increase in r for Lenders & Borrowers

- › Lenders make money on interest
 - Increase in r means they have more lifetime wealth
- › Borrowers have to pay interest
 - Increase in r means they have less lifetime wealth
- › Reverse is true for a fall in r

Increase in r for Lenders

- › Lender is on a higher IC
 - More utility
- › Future consumption rose
- › Current consumption...?

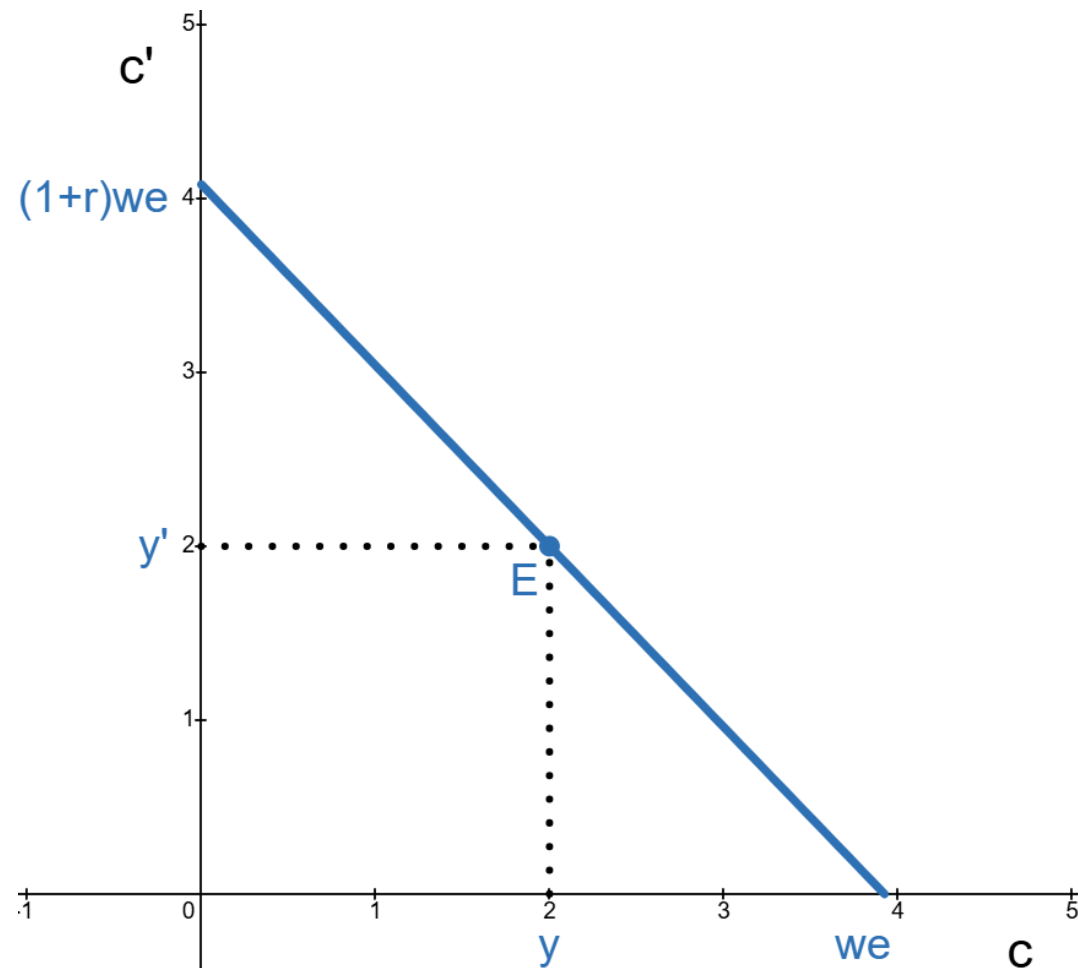


Wealth and Substitution Effects (again)

- › Cannot say what happened to current consumption (c)
- › **Wealth effect:** lender has more lifetime income → consume more c and c'
- › **Substitution effect:** relative price of c has risen → consume more c' and less c
 - Substitute away from relatively expensive good

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Increase in r for Borrowers



For an Increase in r

- › HH's borrow less and lend more
 - Savings increase
- › The quantity supplied of **loanable funds** increases
 - Resources available to be loaned out

Lending to Firms

- › So far we looked at HH's lending to other HH's
- › Now we'll include firms in the model

Why do Firms Borrow?

- › Firms borrow in order to fund investment
 - Ch. 3: investment in capital goods, not the stock market
- › Capital **depreciates** with use so they have to update it
 - Rather a brand new computer or one that's been used for years?
- › Depreciation rate is denoted by δ
 - Depreciated capital is then: δk
 - Roughly 7%/year in aggregate

A Timing Assumption

- › Our variable k represents total capital today, but...
- › Investment will not change k
 - Building a new machine does not happen instantly
 - There is time before you see returns on majority of investments
- › Investment will change k'
 - “one-period time-to-build”
 - Current MPk doesn't matter; future MPk' matters

Capital Accumulation

- › Remaining capital after period: $(1-\delta)k$
- › Capital for next period: k'
- › Investment: i

- › Investment required to get to k' :

$$i = k' - (1-\delta)k$$

- Difference between where you are and where you want to be

The Demand for k'

- › Firms invest so that $MB = MC$
- › Equilibrium condition here is:

$$MPk' = (r + \delta)$$

- › **Marginal benefit:** MPk'
 - Firms invest for the future, so we're concerned with k'
- › **Marginal cost:** price of borrowing + loss via depreciation
 - This is the real **user cost** of a unit of capital

One Final Adjustment for Graphing

- › Usually, we work with prices on the y-axis
- › Conveniently, we can subtract over δ :

$$MPk' - \delta = r$$

- › Remember, r is a relative price like w
 - Recall this is a representative firm so $k = K$
 - Either notation is acceptable

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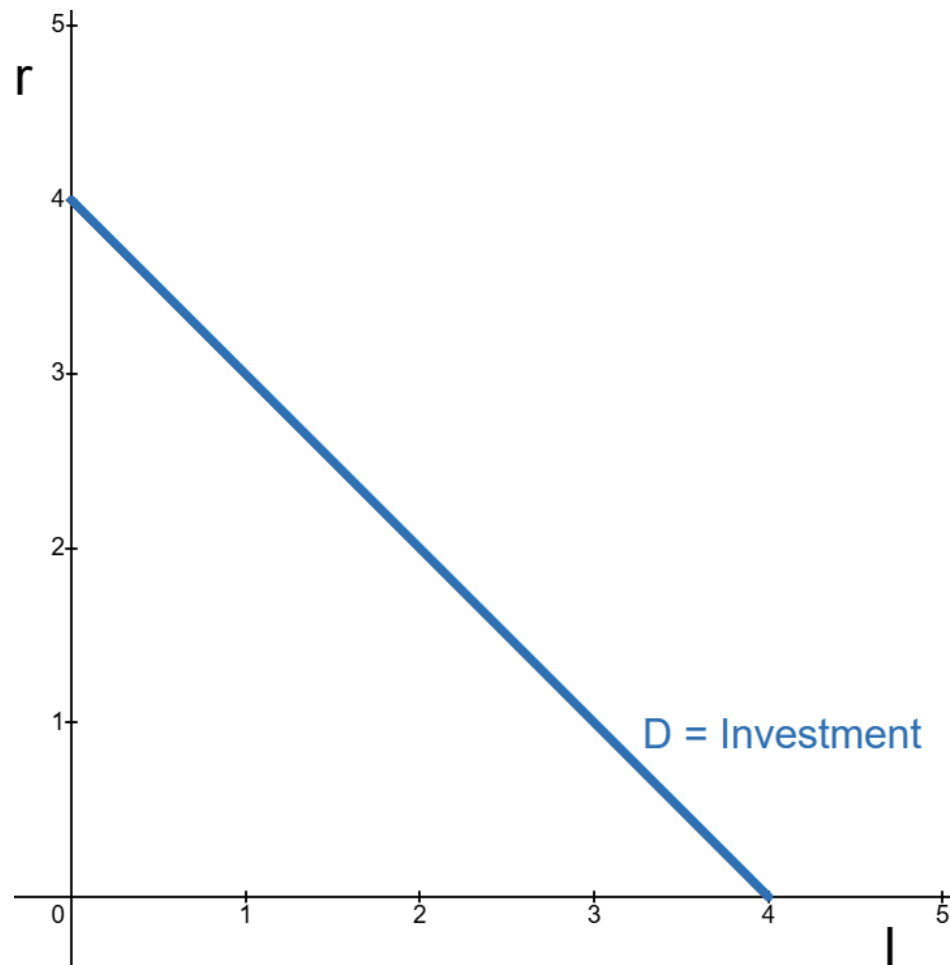
Proof (Not required)

Relationship b/t r and Investment Demand

- › Firms borrow to invest
 - The price of borrowing is r
- › Before we had MPN & w , now have MPk' & r
 - If w is higher than MPN , firms don't hire
 - Now, if r is higher than $MPk' - \delta$, firms don't invest
 - Don't forget δ ; firms don't want to pay for capital that they'll lose

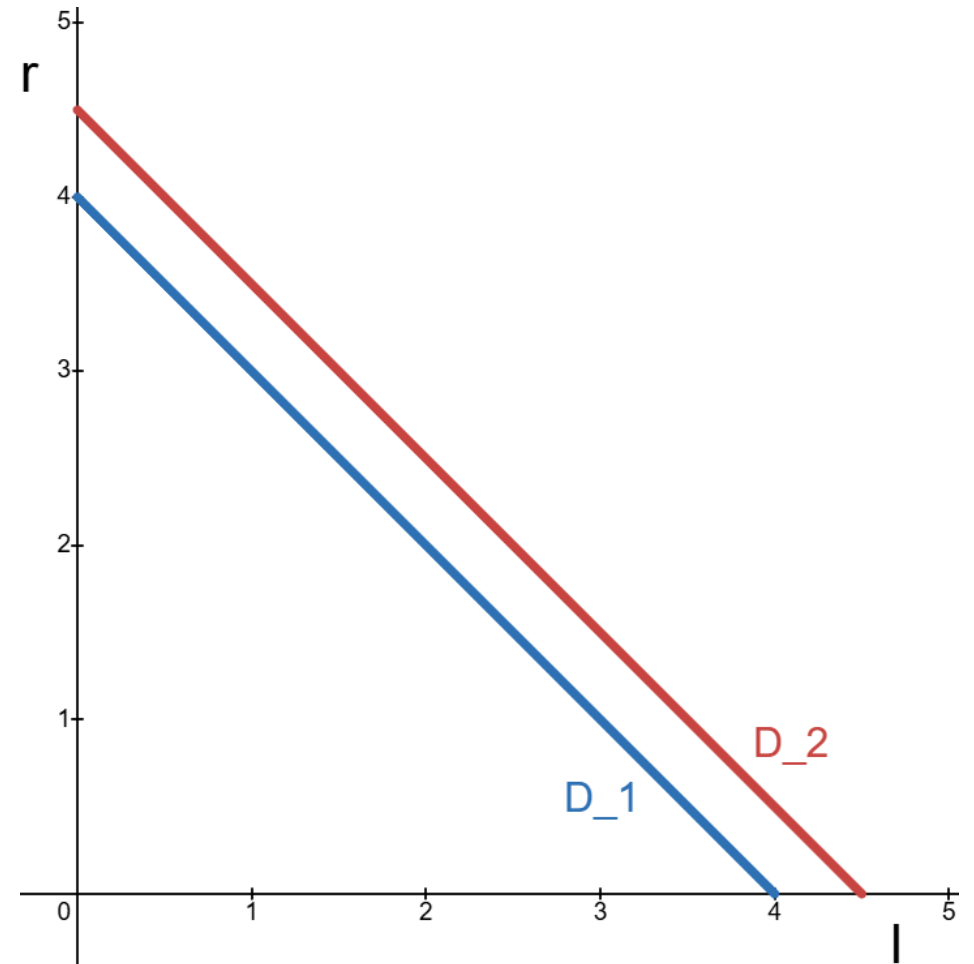
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Firm Investment Demand



Shifts of Demand Curve

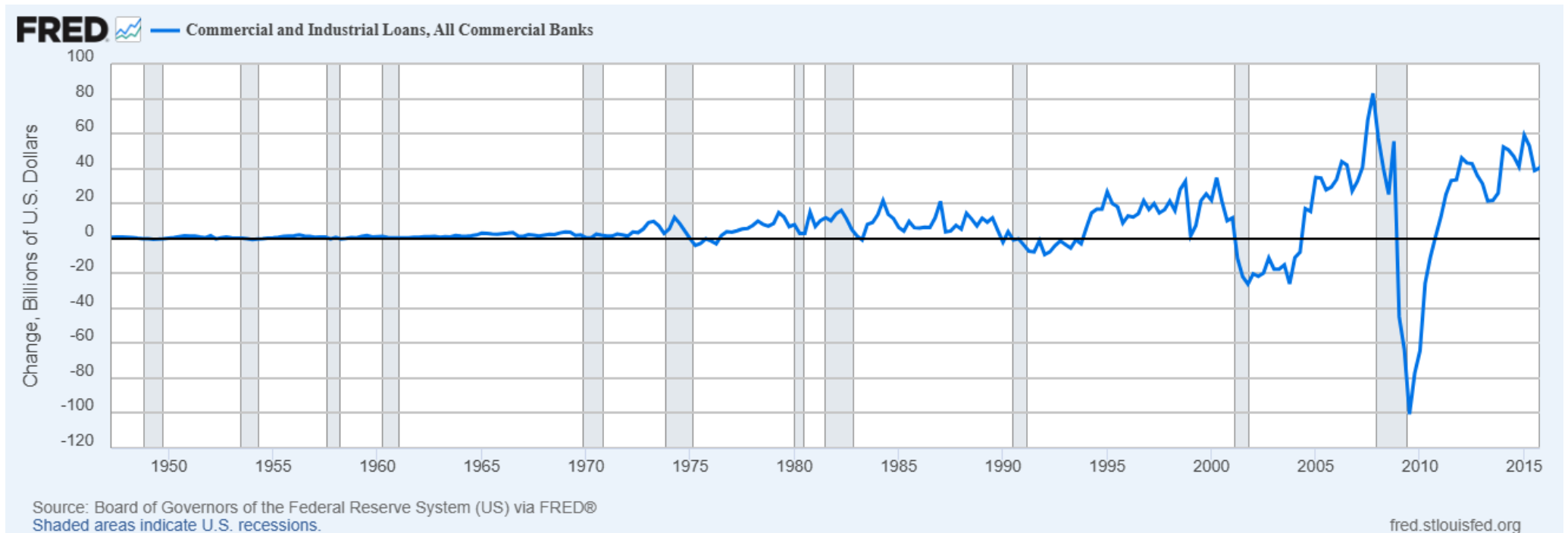
- › Shift right if:
 - MPk' rises (z' or n' rises)
 - k falls
 - δ falls



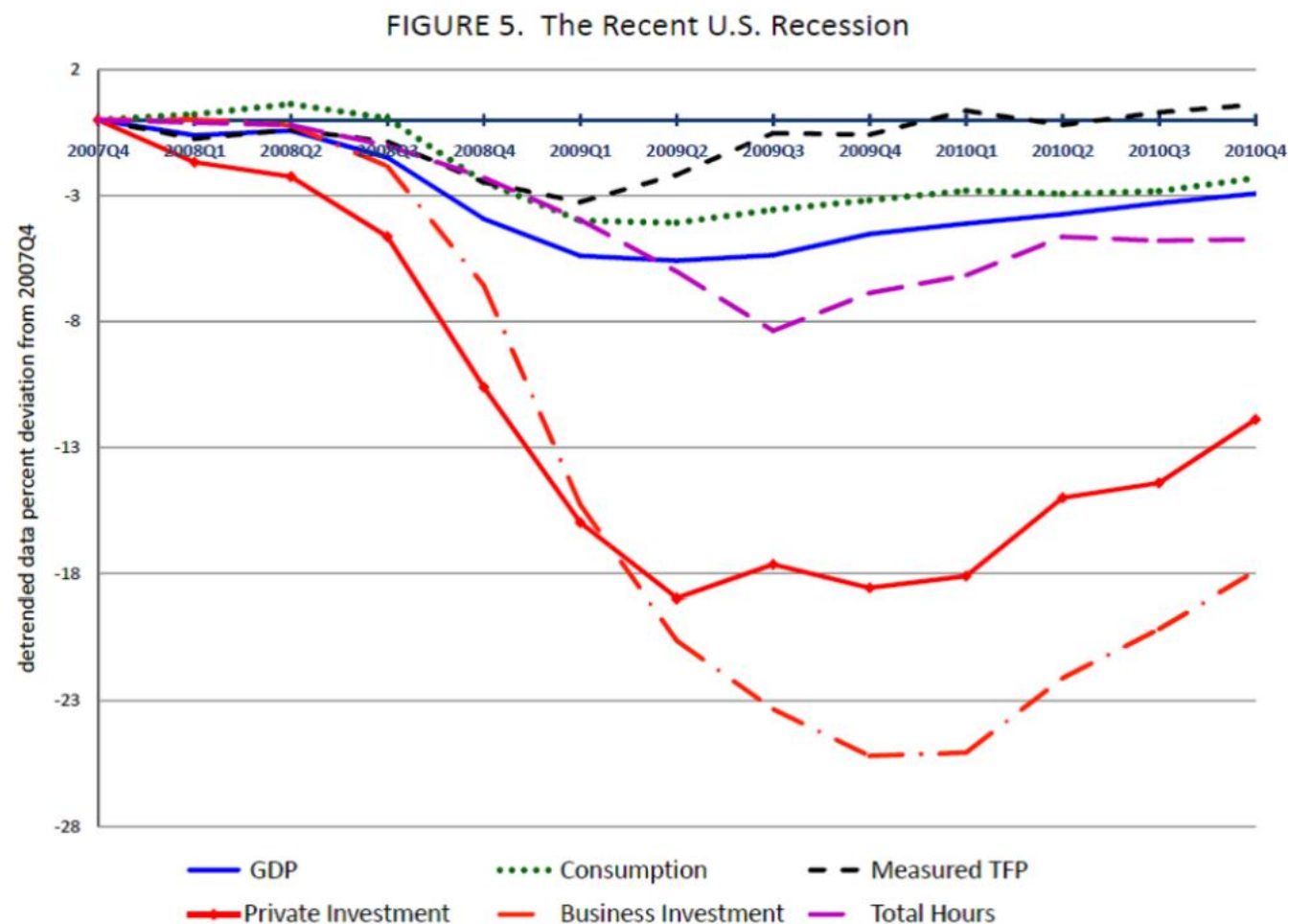
Assumptions?

- › What assumption have we made that is so fundamentally obvious that we may have overlooked it?

C&I Loans: Change, Billions of U.S. Dollars



Borrowed Funds → Investment



(Khan and Thomas, 2013. Figure 5)

Savings and Investment

- › Investment *must* come from savings
 - In fact, in a **closed economy**, *savings = investment*
 - › Closed economy = without international trade
 - Can't invest what's been consumed, so you can only invest from your savings
- › We can see that by looking at Private and Gov't Savings

Private & Government Savings

- › **Private savings:** $S_{pvt} = Y - C + Tr + Int - T$
 - Income (Y)
 - Consumption (C)
 - Transfers from gov't [ex: Social Security] (TR)
 - Interest on gov't debt (INT)
 - Taxes paid (T)

- › **Government (public) savings:** $S_{gov} = T - G - Tr - Int$
 - Taxes collected (T)
 - Government purchases (G)
 - Transfers to households (Tr)
 - Interest payments on debt (Int)

Supply of Savings in a Closed Economy

› Total savings: $S = S_{pvt} + S_{gov}$

› $S = (Y - C + TR + INT - T) + (T - G - TR - INT)$

› Supply of Savings is:

$$S = Y - C - G$$

The Goods Market and $S = I$

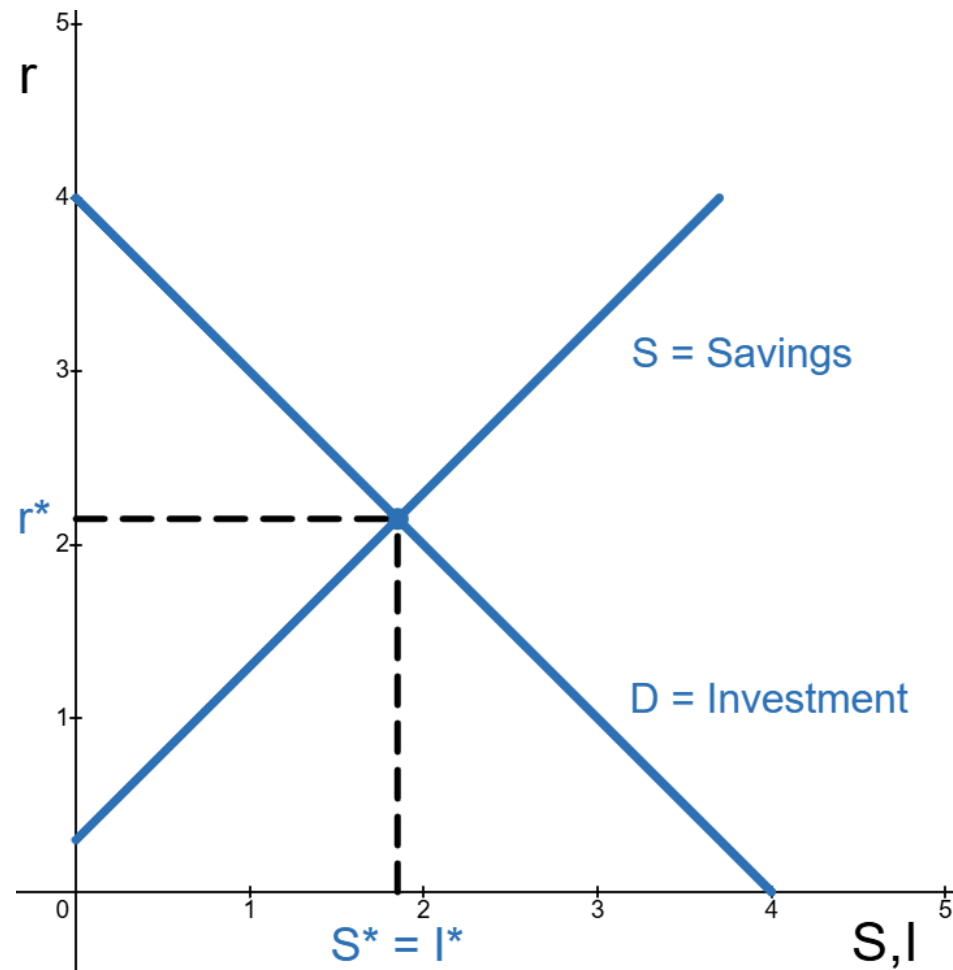
- › From last slide: $S = Y - C - G$
- › Market clearing in Goods Market: $Y = C + I + G \pm \cancel{NX}$
 - $NX = 0$ in closed economy
- › $S = (C + I + G) - C - G$
- **$S = I$**

The Loanable Funds Market

- › In a closed economy, investment must come from savings
- › The **loanable funds market** is where HHs provide savings to be used for investment
 - Supply: HH (net) savings
 - Demand: Firm investment
 - Price: r determines the quantities supplied/demanded
- › Market clearing is an interest rate where quantity demanded = quantity supplied

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Equilibrium



Equilibrium in the Data

Sector	\$ Net Lending (+) / Net Borrowing (-) billions
Domestic businesses	768.032
Households and institutions	563.766
Federal government	-1,958.199
State and local government	-357.520
Rest of world	983.921

Current taxes and transfer payments to the rest of the world: Net lending or net borrowing (-) (BEA, 2026).

❖ Statistical discrepancy is assumed to be 0.

Shifts in Savings Supply

1. Changes to HH wealth will shift the supply curve
 - Increasing current income shifts supply curve right
 - Increasing future income shifts supply curve left
 - Why?

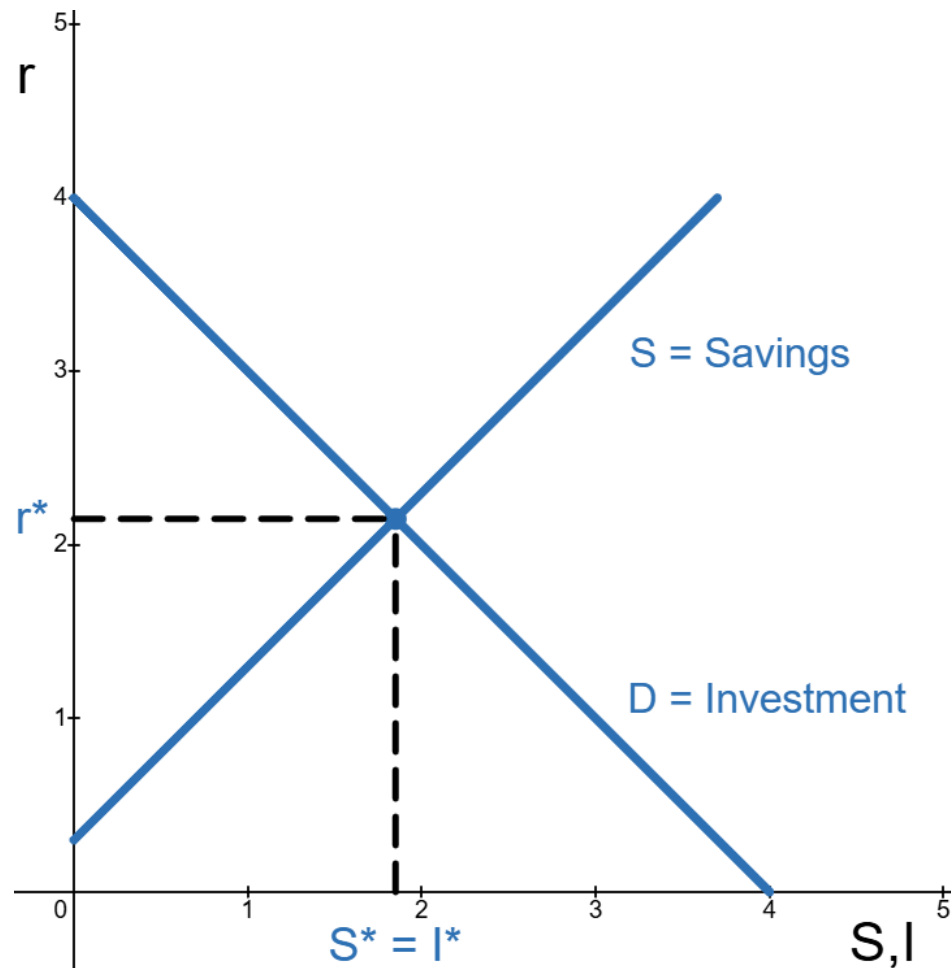
2. Changes to Gov spending also shift the supply curve

Effect of Government Spending

- › HH saving decision is micro-founded
 - Supply of Private Savings
- › No real standard approach for Government Savings
 - Gov acts as its own “aggregate agent” anyway
- › Can just directly look at increase in G
 - Decrease in Government Saving

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Increase in G ; Fall in S_{gov}



References

1. Board of Governors of the Federal Reserve System (US), Commercial and Industrial Loans, All Commercial Banks [BUSLOANS], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/BUSLOANS>, February 2, 2026.
 2. Khan, Aubhik, and Julia K. Thomas. "Credit shocks and aggregate fluctuations in an economy with production heterogeneity." *Journal of Political Economy* 121.6 (2013): 1055-1107.
 3. U.S. Bureau of Economic Analysis, Current taxes and transfer payments to the rest of the world: Net lending or net borrowing (-), NIPAs [W162RC1Q027SBEA], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/W162RC1Q027SBEA>, January 31, 2026.
- A. All non-cited graphs for this course made by me on [Desmos](#).

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